Dynamical Systems Homework #2

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Problem (Exercise 1). *Let $\alpha \in \mathbb{R}$ and $\varepsilon > 0$. Show that there exists $n \in \mathbb{N}$ such that $\|n^3\alpha\| < \varepsilon$ (recall that $\|x\| := \min_{a\in\mathbb{Z}} |a - x|$.)*

Solution. Here, we mimic the proof from the notes (where this result was shown for $\|n\alpha\|, \|n^2\alpha\| < \varepsilon$). Consider the skew rotation (X, T) for $X = \mathbb{R}^3/\mathbb{Z}^3$ and

$$T : X \to X,\ (x, y, z) \mapsto (x + \alpha, y + x, z + y).$$

One verifies that the nth iterate of T is given by:

$$T^n(x, y, z) = \left(x + n\alpha, y + nx + \binom{n}{2}\alpha, z + ny + \binom{n}{2}x + \binom{n}{3}\alpha\right).$$

In particular, what interests us most, $T^n(0, 0, 0) = (n\alpha, \binom{n}{2}\alpha, \binom{n}{3}\alpha)$. The point $(0, 0, 0)$ is recurrent for the same reason as in the notes: one notices that $(x, y, z) \mapsto x$ is a factor map from (X, T) onto $(\mathbb{R}/\mathbb{Z}, x \mapsto x + \alpha)$, and maps $\phi_{\tau,\nu} : (x, y, z) \mapsto (x, y + \tau, z + \nu)$ for $\tau, \nu \in \mathbb{R}/\mathbb{Z}$ are automorphisms of (X, T), showing that every point in (X, T) is recurrent.

Consequently, for every $\varepsilon > 0$ there exists n such that $\|\binom{n}{k}\alpha\| < \frac{1}{13}\varepsilon$ for $k = 0, 1, 2$. Now we notice the following key relation:

$$n^3 = 6\binom{n}{3} + 6\binom{n}{2} + n.$$

This allows us to immediately establish $\|n^3\alpha\| \le 6\|\binom{n}{3}\alpha\| + 6\|\binom{n}{2}\alpha\| + \|n\alpha\| < \varepsilon$.

Problem (Exercise 2). *Consider the full one-sided binary shift $(\{0, 1\}^{\mathbb{N}}, \sigma)$. Construct a point $x \in \{0, 1\}^{\mathbb{N}}$ that is recurrent but not uniformly recurrent.*

Solution. We define the point x as follows: let $x_1 = 1$ and $x_{n+1} = x_n \parallel 0^{n+1} \parallel x_n$ for each $n \in \mathbb{N}$. Then, let x be the limit of $(x_n)_{n\ge0}$. So x begins as

$$x = 1010010100010100101\ldots$$

In $(\{0, 1\}^{\mathbb{N}}, \sigma)$, the point x is recurrent if every finite block appears in x infinitely many times. Take some block b in x, then it must fully lie inside x_k for some large enough k. Since x_{k+1} contains two copies of x_k, and then x_{k+2} contains two copies of x_{k+1} (and thus four copies of x_k), clearly by induction x_k appears infinitely many times in x. Since $b \subseteq x_k$, same holds for b, and thus x is recurrent.

However, since we introduce increasing gaps 0^{n+1} of zeroes between x_k's, this point is not uniformly recurrent. More specifically, one can clearly see this for a block $b = 1$. The distance between the last 1 of the first x_n and the first 1 of the second x_n in the block x_{n+1} is exactly n, which can be arbitrary large as $n \to \infty$.

Problem (Exercise 3). *Let (X, T) be the binary adding machine. Verify that T is a homeomorphism, and show that (X, T) is a minimal topological dynamical system.*

Solution. Recall the definition: binary adding machine is a dynamical system $(\{0,1\}^{\mathbb{N}}, T)$ where T is adding 1 with a carry:

$$T : \{0,1\}^{\mathbb{N}} \to \{0,1\}^{\mathbb{N}}, \qquad \begin{aligned}(1,1,\dots,1,0,x_{k+1},\dots) &\mapsto (0,0,\dots,0,1,x_{k+1},\dots),\\ (1,1,1,\dots) &\mapsto (0,0,0,\dots).\end{aligned}$$

First, we verify that T is a homeomorphism. This map is clearly bijective: the inverse map T^{-1} is defined as follows:

$$T^{-1} : \{0,1\}^{\mathbb{N}} \to \{0,1\}^{\mathbb{N}}, \qquad \begin{aligned}(0,0,\dots,0,1,x_{k+1},\dots) &\mapsto (1,1,\dots,1,0,x_{k+1},\dots),\\ (0,0,0,\dots) &\mapsto (1,1,1,\dots).\end{aligned}$$

Next we verify that both T and T^{-1} are continuous. It suffices to check preimage on basic cylinders $C(a_1,\dots,a_k) := \{x \in \{0,1\}^{\mathbb{N}} : x_1 = a_1,\dots,x_k = a_k\}$. Then,

$$T^{-1}(C(a_1,\dots,a_k)) = \bigcup_{\substack{1\le r\le k\\ (a_i)_{i<r}=0,\, a_r=1}} C(\underbrace{1,\dots,1}_{(r-1)\text{ times}},0,a_{r+1},\dots,a_k),$$

which is a finite union of basic cylinders*. Thus, the preimage of T for every basis open set is open. The construction for T^{-1} is similar.

Now we show minimality, which is equivalent to showing that $\mathcal{O}(x) = \{T^n(x)\}_{n\ge 0}$ is dense in $\{0,1\}^{\mathbb{N}}$ for every $x \in \{0,1\}^{\mathbb{N}}$. Take any x and cylinder $C(a_1,\dots,a_k)$. We need to find n such that $T^n(x) \in C(a_1,\dots,a_k)$. But note that the first k coordinates of $T^n(x)$ cycle through all binary words of length k as n runs through $0,\dots,2^k-1$:

$$\{(T^n(x)_1,\dots,T^n(x)_k) : 0 \le n < 2^k\} = \{0,1\}^k.$$

Therefore we are guaranteed to find some $n < 2^k$ such that $T^n(x) \in C(a_1,\dots,a_k)$. This shows that $\mathcal{O}(x)$ is dense and thus $(\{0,1\}^{\mathbb{N}}, T)$ is minimal.

Problem (Exercise 4). *Let $(a_n)_{n\in\mathbb{N}}$ be a real-valued sequence such that $a_n \ge 0$ and $a_{n+m} \le a_n + a_m$ for all $n, m \ge 1$. Show that*

$$\lim_{n\to\infty} \frac{1}{n} a_n = \inf_{n\in\mathbb{N}} \frac{1}{n} a_n.$$

(In particular, the limit exists and is finite.)

Solution. We show two directions. Further let $\alpha := \inf_{n\in\mathbb{N}} \frac{1}{n}a_n$.

($\ge$) This direction is easy. By definition of infinum, $\frac{1}{n}a_n \ge \alpha$ for all n. Therefore, $\liminf_{n\to\infty} \frac{1}{n}a_n \ge \alpha$.

($\le$) Fix $\varepsilon > 0$. By definition of infinum, there exists $k \in \mathbb{N}$ such that $\frac{1}{k}a_k < \alpha + \varepsilon$. Divide n by k and write $n = qk + r$ (for $0 \le r < k$). Using condition $a_{n+m} \le a_n + a_m$, we obtain $a_n \le q a_k + a_r$. Divide by n and bound expression given $qk/n \le 1$:

$$\frac{1}{n}a_n \le \frac{qk}{n}\cdot\frac{a_k}{k} + \frac{a_r}{n} \le \frac{a_k}{k} + \frac{C}{n}, \quad \text{where } C := \max_{0\le r<k} a_r.$$

*To be fully correct, one should also include $C(1,\dots,1)$ in the preimage if all $a_i = 0$.

By taking limit as $n \to \infty$, we get

$$\limsup_{n\to\infty} \frac{1}{n} a_n \leq \frac{a_k}{k} < \alpha + \varepsilon.$$

Since ε was arbitrary, $\limsup_{n\to\infty} \frac{1}{n} a_n \leq \alpha$. Combining two directions, we obtain

$$\alpha \leq \liminf_{n\to\infty} \frac{1}{n} a_n \leq \limsup_{n\to\infty} \frac{1}{n} a_n \leq \alpha,$$

implying $\lim_{n\to\infty} \frac{1}{n} a_n = \alpha$.

Problem (Exercise 5). *Let $C \in \mathbb{N}$ and let $T : \mathbb{R}/\mathbb{Z} \to \mathbb{R}/\mathbb{Z}$ be a map with the property that for each $y \in \mathbb{R}/\mathbb{Z}$ there are at most C values of X such that $T(x) = y$. Show that $h(\mathbb{R}/\mathbb{Z}, T) \leq \log C$. Find an example where $h(\mathbb{R}/\mathbb{Z}, T) = \log C$.*

Solution. We will use the definition of entropy through (n, ε)-separated sets. Specifically, if $\operatorname{maxsep}(n, \varepsilon)$ denotes the size of the maximum (n, ε)-separated set (*i.e.*, $d_n(x, y) := \max_{0 \leq k < n} d(T^k(x), T^k(y)) \geq \varepsilon$ for all x, y in such set), then

$$h(\mathbb{R}/\mathbb{Z}, T) = \lim_{\varepsilon \to 0} \limsup_{n\to\infty} \frac{1}{n} \log \operatorname{maxsep}(n, \varepsilon).$$

Further denote $X := \mathbb{R}/\mathbb{Z}$. Fix $n \in \mathbb{N}$, $\varepsilon > 0$, and let $X = \bigcup_{i=1}^{K} X_i$ be the partition of X into $K = K(\varepsilon)$ intervals $(X_1, \ldots, X_K)$ of diameter below ε (such partition is finite since X is compact). Define the map $\iota_{n,\varepsilon} : X \to \Sigma_K^n$ (where $\Sigma_K := \{1, \ldots, K\}$) as follows: $\iota_{n,\varepsilon}(x)_k$ is the index of the interval where $T^k(x)$ lands ($0 \leq k < n$).

Proposition 1. $\operatorname{maxsep}(n, \varepsilon) \leq \#\iota_{n,\varepsilon}(X)$.

Indeed, if $\iota_{n,\varepsilon}(x) = \iota_{n,\varepsilon}(y)$ for $x \neq y \in X$, then x, y are not (n, ε)-separated.

Proposition 2. $\#\iota_{n,\varepsilon}(X) \leq K(\varepsilon) C^{n-1}$.

Fix a sequence $(i_0, i_1, \ldots, i_{n-1}) \in \Sigma_K^n$. Pick some point z_{n-1} such that $T^{n-1}(z_{n-1}) \in X_{i_{n-1}}$. Now we need some point z_{n-2} such that $T(z_{n-2}) = z_{n-1}$. According to the problem, the number of choices is at most C. Consequently, the number of choices for z_{n-3} is at most C^2. After n backward steps, there are at most C^{n-1} possible initial points x such that $\iota_{n,\varepsilon}(x) = (i_0, i_1, \ldots, i_{n-1})$. Since we did not control the choice of the initial cell i_{n-1}, we multiply C^{n-1} by $K(\varepsilon)$.

Using two propositions, we obtain

$$\operatorname{maxsep}(n, \varepsilon) \leq K(\varepsilon) C^{n-1}.$$

Apply $\limsup_{n\to\infty} \frac{1}{n} \log(\cdot)$ to both sides:

$$\limsup_{n\to\infty} \frac{1}{n} \log \operatorname{maxsep}(n, \varepsilon) \leq \limsup_{n\to\infty} \frac{\log K(\varepsilon)}{n} + \limsup_{n\to\infty} \frac{n-1}{n} \log C = \log C$$

Taking $\varepsilon \to 0$ we finally obtain $h(X, T) \leq \log C$.

Example when $h(X, T) = \log C$. Take $(\mathbb{R}/\mathbb{Z}, x \mapsto Cx)$. Each point has exactly C preimages and its entropy is $\log C$ (which was proven during our classes).

[Vri14] J. Vries. *Topological Dynamical Systems: An Introduction to the Dynamics of Continuous Mappings*. De Gruyter Studies in Mathematics. De Gruyter, 2014. URL: https://books.google.com.ua/books?id=hyroBQAAQBAJ.